

Lab 8: Slotted Line Lab and VSWR Measurements

ETE 4371

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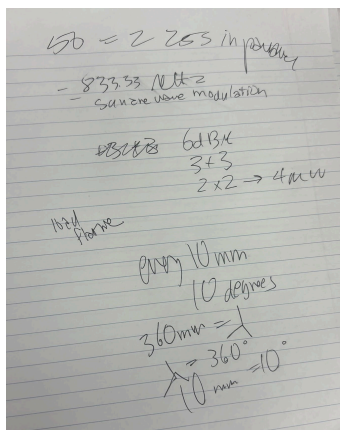
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Mariana Rivera

Introduction

V_{max} and V_{min} are relatively easy to measure, so we can use the VSWR to assess how much of a match we have. VSWR (voltage standing ratio) essentially tells us how bad the reflections are on a transmission line, or how bumpy the voltage pattern is along the cable because of reflected waves. It can be calculated as $VSWR = \frac{V_{max}}{V_{min}} = \frac{\text{largest voltage on the standing wave}}{\text{smallest voltage on the standing wave}}$ and $VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}$. If a signal travels down a transmission line towards the load and the load matches the line, then the signal gets absorbed by the load and there is no reflection coming back. In this case, $VSWR=1$. However, if the load is not a match, then parts of the signal reflect back and the forward wave and reflected wave overlap, creating spots with more and less amounts of voltage. In this case, $VSWR > 1$, which is not desirable.

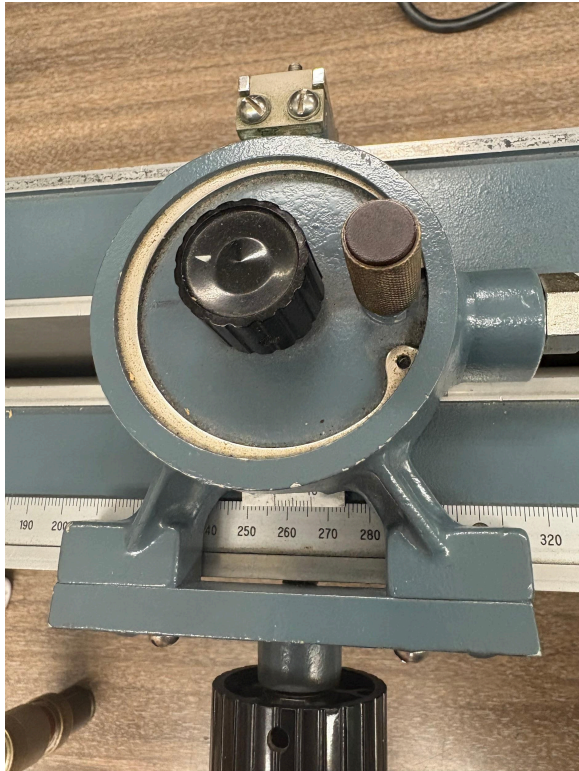
From our previous labs, we know that there is a standing wave within a transmission line. We also know that it is possible to measure this standing wave and the phase of it, although this can be tricky. For instance, any physical connection to the transmission line can change the standing wave. How then can we measure the standing wave? In this lab, we will be doing just that by using a slotted transmission line. This way, the line uses a movable probe that doesn't touch the sides of the transmission line in order to measure the standing wave.



Img. 1: Paper calculations on the relationship between phase in degrees and 833.33MHz

Data

Using the slotted transmission line, we were able to determine values for the standing wave when the terminating load is an open, a short, a match, 25 ohms, and 200 ohms. The results are recorded and plotted accordingly.



Img. 2: Probe Setup



Img. 3: Terminating load

Match

Sample Data when the load is 50 Ohms.

Position (mm)	Voltage (mV)
20	922.00
30	920.00
40	912.29
50	925.35
60	918.93

70	927.98
80	911.13
90	909.48
100	889.03
110	854.12
120	877.63
130	927.07
140	916.04
150	916.24
160	912.96
170	907.37
180	899.66
190	901.51
200	894.49
210	907.31
220	891.34
230	895.36
240	901.63
250	883.75
260	870.76
270	873.64
280	874.44
290	884.86
300	886.18
310	896.99
320	885.50
330	902.05

340	894.70
350	895.97
360	888.55
370	870.87
380	860.06
390	876.39
400	865.02

Table 1.

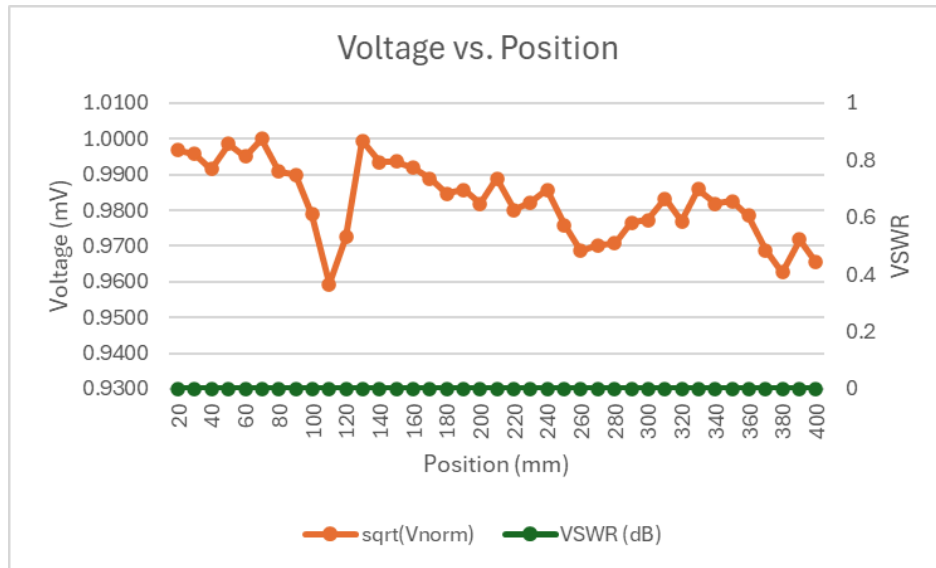
Find the peaks.

After finding the peaks, record data at the positions where the peaks are.

Position (mm)	Voltage	Sqrt (Voltage)	Probe/Voltage Condition
0	0.942	0.971	constant
100	0.907	0.952	constant
200	0.933	0.966	constant
300	0.933	0.966	constant

Table 2.

Now we will normalize the data. How will we do that? We will find the max value, divide everything by that value, then take the square root of normalized data, and plot that versus position.



Img 4.

Open

Sample Data when the load is Open.

Position (mm)	Voltage (mV)
20	431.78
30	584.9
40	725.6
50	833.3
60	902.5
70	916.51
80	886.39
90	812.87
100	697.26
110	552.93
120	394.70
130	240.33
140	111.02

150	26.35
160	3.21
170	33.1
180	124.2
190	258.7
200	412.5
210	572.3
220	714.8
230	826.0
240	897.5
250	920.0
260	892.2
270	819.0
280	705.7
290	552.3
300	401.0
310	249.0
320	119.5
330	31.9
340	3.20
350	28.8
360	115.0
370	243.5
380	380.1
390	555.0
400	696.0

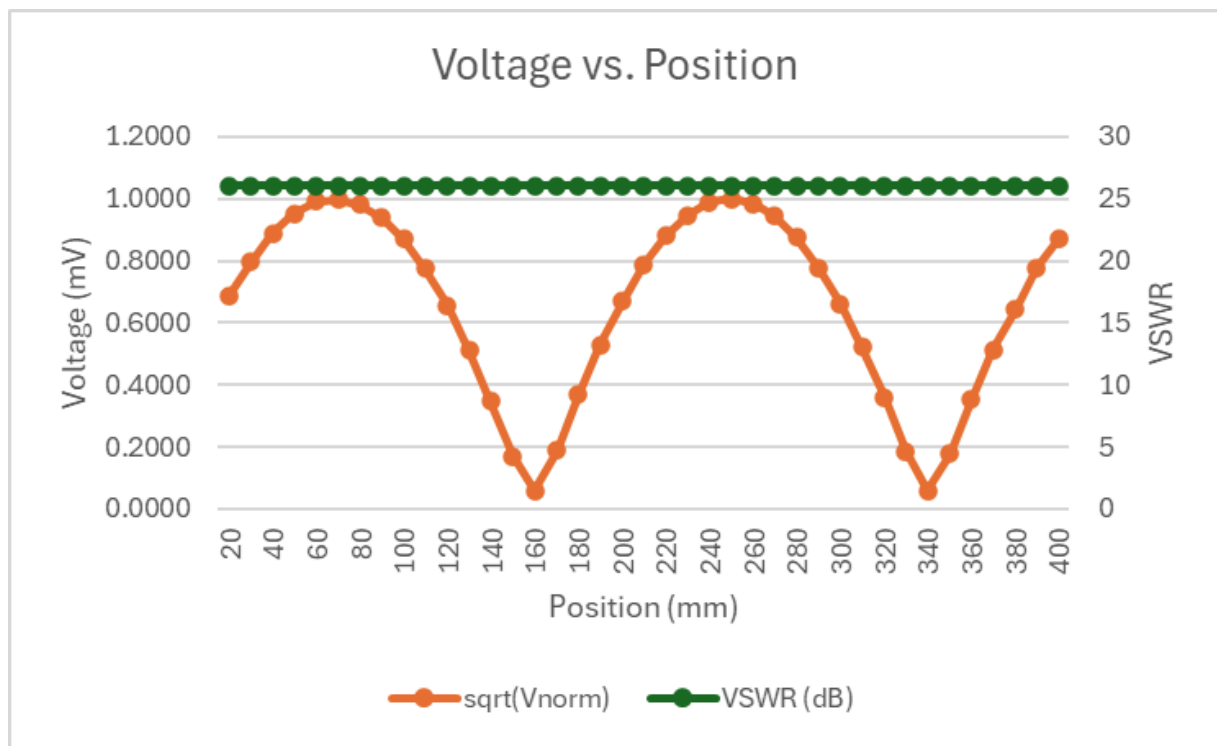
Table 3.

Find the peaks

After finding the peaks, record data at the positions where the peaks are.

Position (mm)	Voltage (mV)	Sqrt (Voltage)	Probe/Voltage Condition
69.4	915.38	30.256	peak
161.6	3.43	1.852	null
256.9	898.5	29.975	peak
336.3	5.02	2.241	null

Table 4.



Img 5.

Short

Sample Data when the load is Short.

Position (mm)	Voltage (mV)
20	997.54
30	909.30

40	735.50
50	504.18
60	264.46
70	75.087
80	3.557
90	91.512
100	304.89
110	512.14
120	758.58
130	903.34
140	966.01
150	964.11
160	930.08
170	923.58
180	956.34
190	999.58
200	996.94
210	871.13
220	690.89
230	456.42
240	204.57
250	38.06
260	4.999
270	140.10
280	375.22
290	600.89
300	810.29

310	930.06
320	975.57
330	941.33
340	933.61
350	948.90
360	1.005 V
370	1.061 V
380	1.029 V
390	875.23
400	688.16

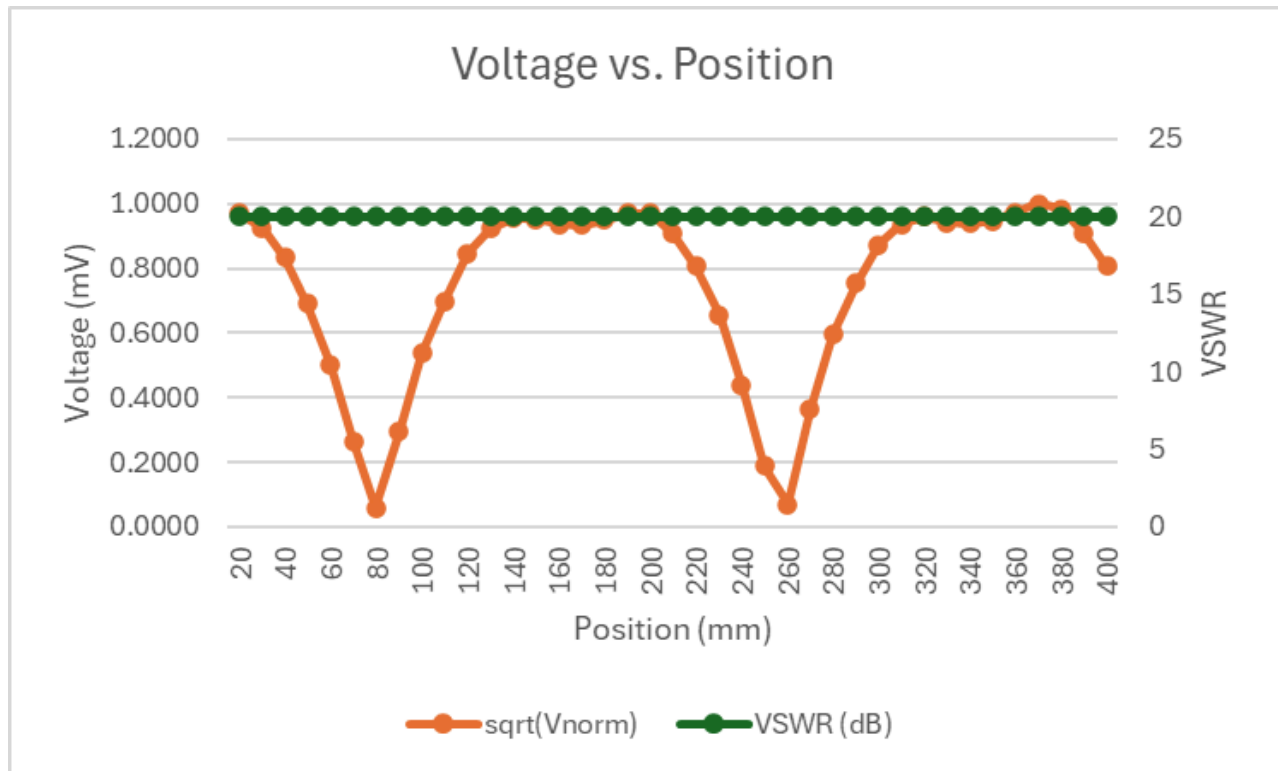
Table 5.

Find the peaks

After finding the peaks, record data at the positions where the peaks are.

Position (mm)	Voltage (mV)	Sqrt (Voltage)	Probe/Voltage Condition
80.4	4.505	2.122	null
190.6	977.59	31.266	peak
256.0	3.602	1.898	null
362.1	988.27	31.437	peak

Table 6.



Img 6.

200 Ohms

Sample Data when the load is 200 Ohms.

Position (mm)	Voltage (mV)
20	230.97
30	285.42
40	398.15
50	562.66
60	738.89
70	881.66
80	930.51
90	903.30
100	765.09
110	602.04

120	433.73
130	307.49
140	242.02
150	233.56
160	239.58
170	247.26
180	238.43
190	225.55
200	244.91
210	316.84
220	445.85
230	620.35
240	795.15
250	923.16
260	958.44
270	888.16
280	733.67
290	577.68
300	396.38
310	287.09
320	234.76
330	235.29
340	255.60
350	255.77
360	242.20
370	227.53
380	255.06

390	339.23
400	486.86

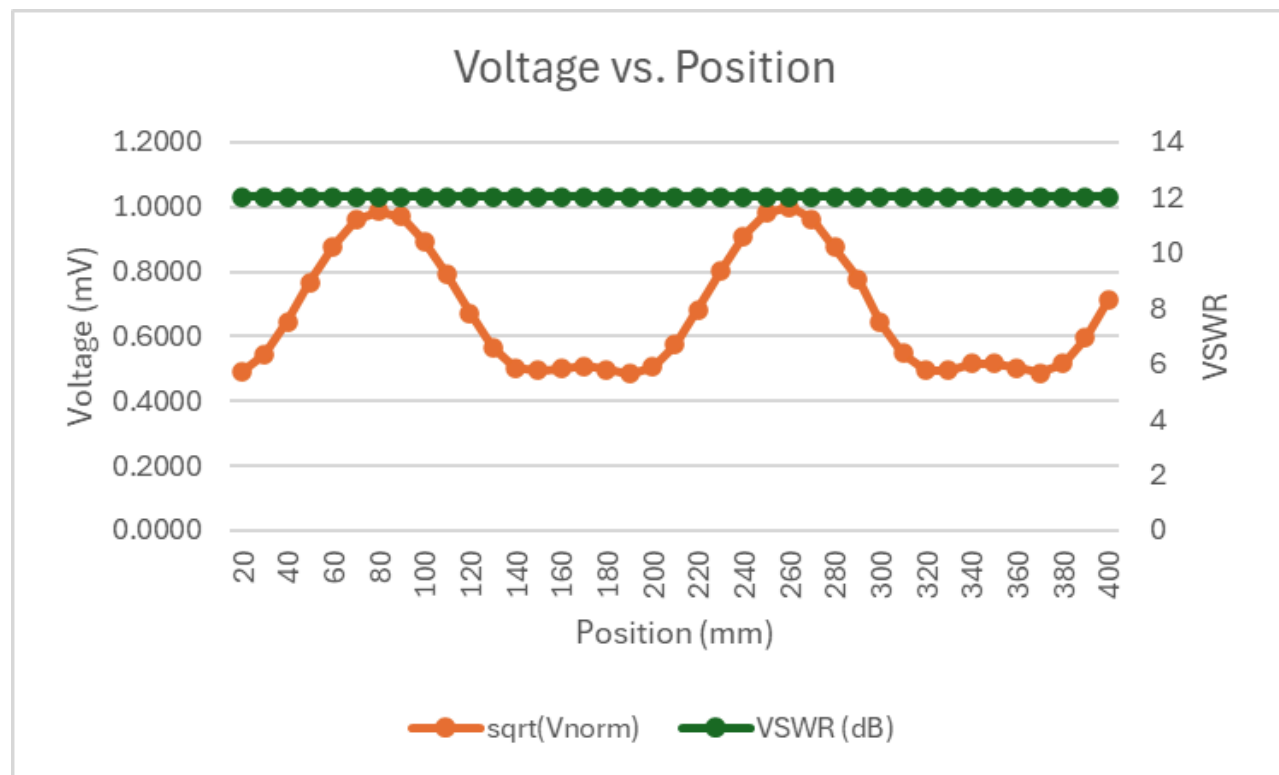
Table 7.

Find the peaks

After finding the peaks, record data at the positions where the peaks are.

Position (mm)	Voltage (mV)	Sqrt (Voltage)	Probe/Voltage Condition
79.7	973.03	31.193	peak
145.3	225.83	15.028	null
252.8	973.86	31.207	peak
322.4	230.57	15.185	null

Table 8.



Img 7.

25 Ohms**Sample Data when the load is 25 Ohms.**

Position (mm)	Voltage (mV)
20	906.66
30	808.35
40	676.44
50	527.70
60	386.63
70	306.31
80	332.40
90	432.40
100	589.99
110	728.25
120	864.99
130	922.34
140	921.55
150	898.52
160	876.04
170	883.35
180	906.23
190	906.67
200	882.61
210	771.97
220	68.49
230	481.18
240	348.90

250	291.51
260	339.85
270	472.80
280	619.10
290	774.24
300	882.28
310	933.19
320	918.90
330	889.23
340	871.21
350	885.49
360	908.40
370	907.57
380	850.80
390	744.42
400	586.32

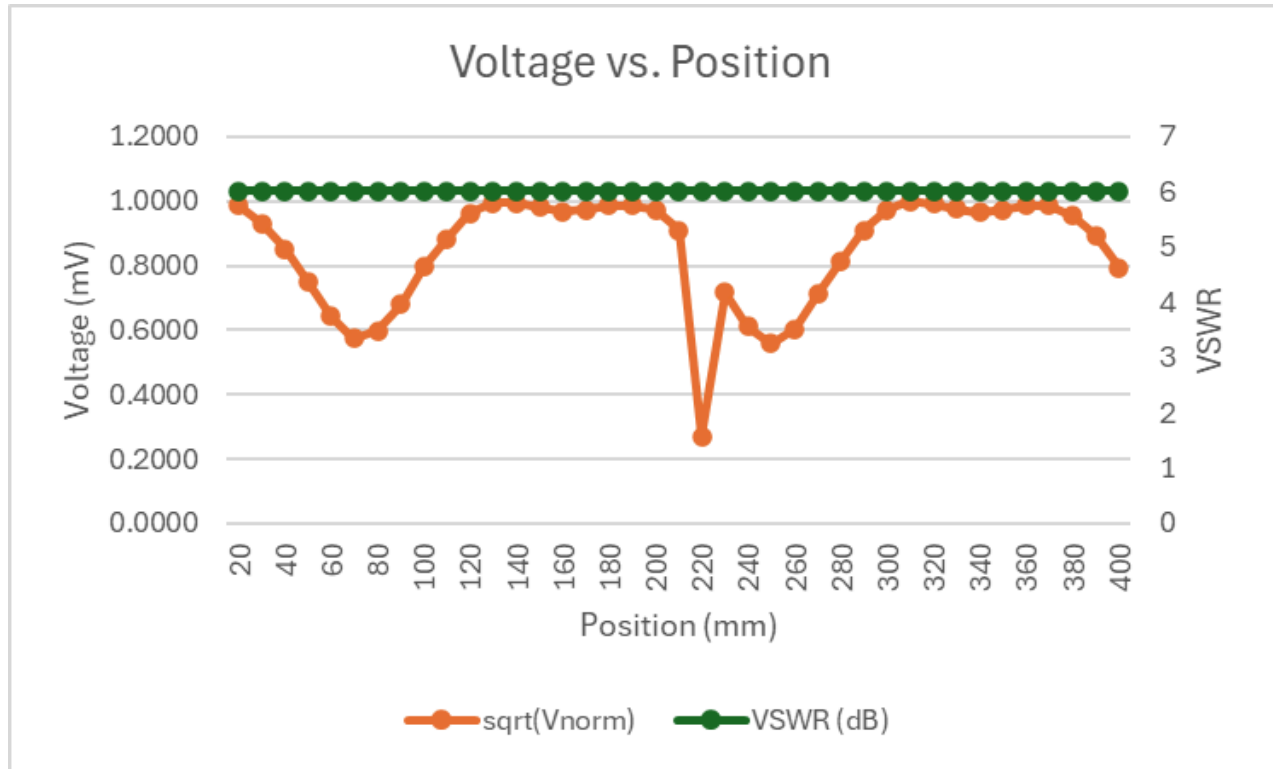
Table 9.

Find the peaks

After finding the peaks, record data at the positions where the peaks are.

Position (mm)	Voltage (mV)	Sqrt (Voltage)	Probe/Voltage Condition
174.3	300.28	17.329	null
134.8	934.22	30.565	peak
250.0	301.97	17.377	null
313.5	958.23	30.955	peak

Table 10.



Img 8.

Conclusion

In this lab, we explored how standing wave form on a transmission line when different terminating loads are connected. The slotted line set up was a unique way for us to measure the RF standing waves effectively without disturbing the line through much, serving as a very useful method of understanding VSWR and the reflection coefficient. Using the slotted transmission line, we measured the voltage pattern along the line for a matched load, open load, short load, 25Ω load and 200Ω load. The matched load produced the most constant voltage pattern, which makes sense because the load impedance matched the characteristic impedance of the line, resulting in little to no reflected wave. In contrast, the open and short loads produced large peaks and places with unproportionately small voltages, caused by the incident wave reflecting back towards the source. The 25Ω load and 200Ω load produced patterns that were between the matched case and open/ short cases. These loads were mismatched, so some reflection occurred but not as extreme as the open and short conditions.

Through comparison of the maximum and minimum voltage values, we were able to relate our measurements to VSWR and see first hand how VSWR indicates how well the load matches. Overall, this lab showed us that transmission line behavior heavily depends on the relationship between the load

impedance and characteristic impedance of the line. When the load is matched, reflections are minimized- if the load is mismatched, reflected waves create peaks and nulls that are extreme enough to be measured and impact the system. This concept is critical to consider when working with transmission lines and understanding reflected signals, thus making the content in this lab important to understand.